

AJITHKUMAR T

Tekla Structures API Developer | C# / .NET Developer
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PROFESSIONAL SUMMARY

Software Developer with 3+ years of experience building **Tekla Structures Open API** automation for structural steel detailing and fabrication, including QA/QC checkers, NC/DSTV/KISS exporters and shop-floor reporting in C# and VB.NET. Architect of **Struzon API Tools**, a ~31,000-line C# fabrication automation suite, and of a **drawing-to-model QA engine** that checks PDF structural drawings against the live Tekla model. Also delivers business modules in PHP/CodeIgniter 3 and administers the company's Linux servers and KVM virtualization, owning work from database schema and backend logic through UI and deployment.

TECHNICAL SKILLS

Engineering APIs	Tekla Structures Open API (2023–2026), Tekla plugins and Akit macros, SDS2 API (Python), AutoCAD/AutoLISP; DSTV/NC, DXF and KISS (.kss) formats
Languages	C#, VB.NET, PHP, Python, JavaScript, SQL
.NET	.NET Framework 4.8, WinForms, WPF, ASP.NET MVC, background STA threading, built-in self-test CLIs
PDF / OCR / Reports	PdfPig, iText 8, Pdfium, Tesseract OCR, NHunspell; Excel Interop, EPPlus, ClosedXML, NPOI, OpenXML, PhpSpreadsheet
Web	CodeIgniter 3, HTML, CSS, Bootstrap, jQuery, AJAX, DataTables, Select2, FullCalendar, Chart.js
Databases	MySQL / MariaDB, PostgreSQL, SQLite, SQL Server
Infrastructure	AlmaLinux, Rocky Linux, Ubuntu, CentOS; KVM/QEMU/libvirt with bridged networking, Docker, Samba, DNS, Active Directory, Mailcow, Mattermost, firewallD
Tools	Git, Visual Studio, VS Code

PROFESSIONAL EXPERIENCE

Software Developer — Struzon Technologies Pvt Ltd

Sep 2023 – Present

Coimbatore, India

Tekla & Engineering Automation

- Architected and built **Struzon API Tools** (~31K lines, ~48 forms) and migrated the full codebase from legacy VB.NET to C# 7.3 on the Tekla 2025 API (see Key Projects).
- Built a **Structural Drawing QA tool** (~9K lines) that compares PDF structural drawings with the live Tekla model for grids, column locations, profiles and plan orientation (see Key Projects).
- Maintained the **KISS/MIS NC exporter** (VB.NET): fixed D-line part-position duplication on channel profiles, sheet-gauge vs. plate classification, and IK circular-contour hole sizing using an empirically verified correction constant.
- Built a **Prefix QA Checker** plugin validating naming/prefix rules; eliminated a UI freeze by moving a blocking Tekla API call to a background STA thread with timeout, and corrected single-part and assembly main-part detection.
- Delivered supporting tools: WPF **Part Data Exporter** (CSV/Excel via ClosedXML), **Imperial Hole Tolerance Checker** (1/16" bolt-hole validation), **GridScheduler** (column grid and bay lengths in imperial format) and a **Bolt Summary** tool pairing a Tekla report template with Excel output.
- Wrote Python scripts against the **SDS2 API** for dimension-text reformatting (Drawing.SearchAndReplace) and BOM customization.

Business Applications — EMS Portal (PHP / CodeIgniter 3)

- Built the **Candidate Referral** module from scratch: 5-table schema, resume parsing with pdftotext and Tesseract OCR, and role-gated access for PM/HR/Admin.
- Built the **RFI Tracker** (cascading client/project filters, AJAX, role-scoped data, Excel export) and replaced hardcoded PM/APM/TL switch-case logic with a database-driven **team-manager mapping** and admin CRUD.
- Extended Conference Room Booking (booked-slot timeline, conflict highlighting, FullCalendar dashboard); automated performance-review reminder emails via cron; fixed invoice filtering/pagination and MySQL schema and FK-constraint issues.

Linux Infrastructure & Virtualization

- Led the rebuild of the internal mail server on **Mailcow (Docker) / Rocky Linux 9** after a ransomware incident, covering DNS/MX records, firewall NAT and a domain-restricted send/receive policy.
- Administer a **KVM hypervisor** (Dell PowerEdge R740, 80 logical CPUs, 125 GB RAM) hosting the CRM server, a Docker/PostgreSQL UAT environment and other VMs; handle capacity planning, bridged networking and VM disk recovery.

- Maintain AlmaLinux **Samba file servers** and the **Mattermost** chat server; upgraded Mattermost from v9.7 to v10.11 ESR through staged releases and enabled Calls screen sharing.

Full-Stack .NET Trainee — Knila IT Solutions

2022 – 2023

Coimbatore, India

- Completed full-stack training in C#, .NET and Angular, building web applications end to end; moved from operations into software development.

Process Executive — Personiv

Feb 2019 – Jul 2022

- Processed large-scale enterprise data on Salesforce and ACRM while meeting SLA, accuracy and compliance targets — foundation for the data-validation mindset behind later QA tooling.

KEY PROJECTS

Struzon API Tools — Steel Fabrication Automation Suite

C#, .NET Framework 4.8, WinForms, Tekla Open API 2025, SDS2, SQLite

- Modular suite of ~48 forms across 3 tool shells converting Tekla and SDS2 exports into shop-ready Excel deliverables: BOMs, buy lists, nesting, shipping lists, bolt summaries, galvanizing and shop-floor reports.
- Direct Tekla API export of DSTV/NC, DXF and KISS (.kss) files from selected objects, with automatic renaming by assembly mark and generated Tekla Akit automation macros.
- PDF drawing toolkit (iText 8, PdfPig) for batch renaming of drawing sets and spell-checking drawing text with NHunspell.
- SQLite tracking layer for part lists and shop operations; machine-locked licensing (CPU ID + AES-256 via BouncyCastle); built-in self-test CLI covering forms, crypto, schema and resources.

Structural Drawing ↔ Tekla Model QA Tool

C#, WinForms (x64), Tekla Open API, PdfPig, Pdfium, Tesseract OCR

- Extraction pipeline merging native PDF text, vector geometry and targeted OCR fallback into a unified drawing model of grids, columns and dimensions, with per-value confidence and source tracking.
- Spatial grid detection and dimension association that parses imperial strings (e.g. 33'-0") to millimetres while preserving the original text for reporting.
- Live Tekla integration: selection sync, vertical-member detection, grid matching and column orientation from the coordinate system; results classified as PASS / FAIL / REVIEW / NOT MATCHED.
- PDF viewer with tiled rendering and toggleable overlays, CSV/debug exports, and 32 automated self-tests.

EDUCATION

Bachelor of Engineering, Electronics and Communication Engineering — 2018

Diploma, Electronics and Communication Engineering — 2014